

International Economic Relations

ECON 308

The Instruments of Trade Policy
(Chapter 9)

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Types of Tariffs

- A tariff is a tax levied when a good is imported.
- A **specific tariff** is levied as a fixed charge for each unit of imported goods.
 - For example, \$3 per barrel of oil.
- An **ad valorem tariff** is levied as a fraction of the value of imported goods.
 - For example, 25% tariff on the value of imported trucks.

Supply, Demand, and Trade in a Single Industry (1 of 4)

- Consider how a tariff affects a single market, say that of wheat.
- Suppose that in the absence of trade the price of wheat is higher in Home than it is in Foreign.
- With trade, wheat will be shipped from Foreign to Home until the price difference is eliminated.

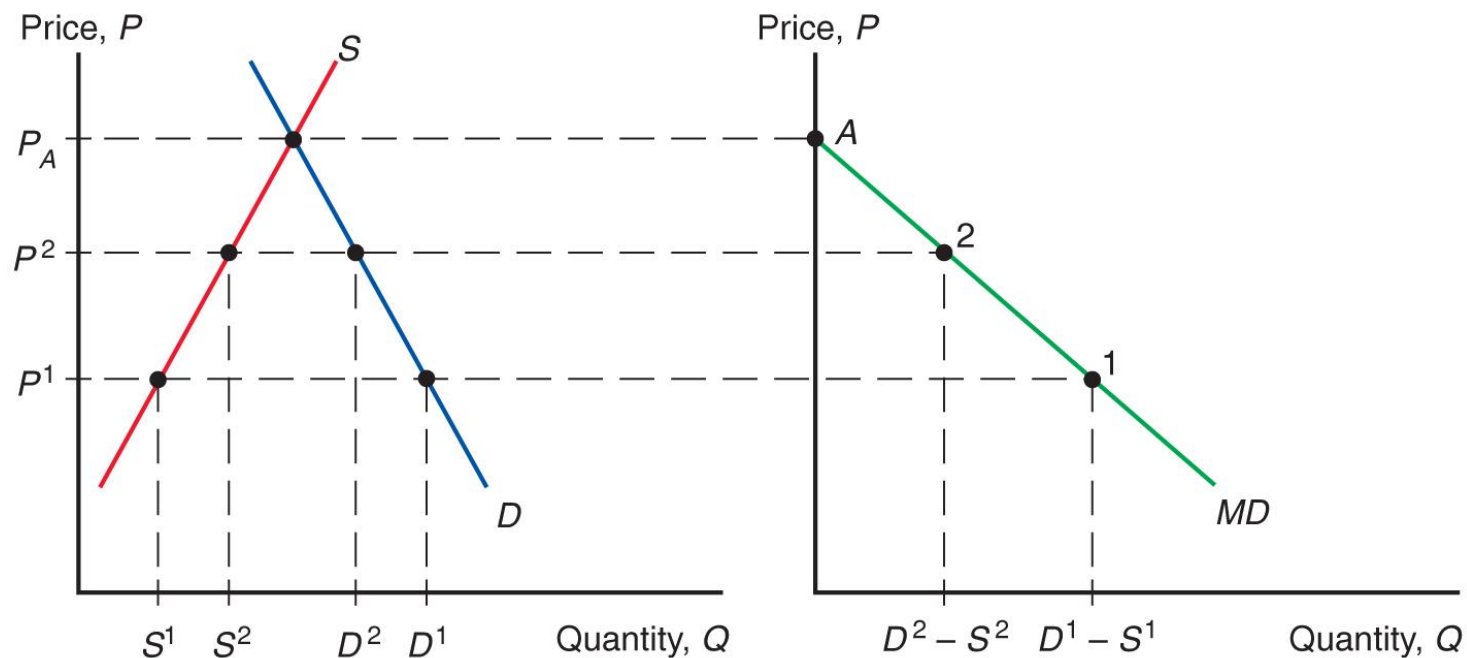
Supply, Demand, and Trade in a Single Industry (2 of 4)

- An **import demand** curve is the difference between the quantity that Home consumers demand minus the quantity that Home producers supply, at each price.
- The Home import demand curve

$$MD = D - S$$

- intercepts the price axis at P_A and is downward sloping:
 - As price increases, the quantity of imports demanded declines.

Figure 9.1 Deriving Home's Import Demand Curve



As the price of the good increases, Home consumers demand less, while Home producers supply more, so that the demand for imports declines.

Supply, Demand, and Trade in a Single Industry (3 of 4)

- An export supply curve is the difference between the quantity that Foreign producers supply minus the quantity that Foreign consumers demand, at each price.
- The Foreign export supply curve

$$XS^* = S^* - D^*$$

- intersects the price axis at P_A^* and is upward sloping:
 - As price increases, the quantity of exports supplied rises.

Figure 9.2 Deriving Foreign's Export Supply Curve



As the price of the good rises, Foreign producers supply more while Foreign consumers demand less, so that the supply available for export rises.

Supply, Demand, and Trade in a Single Industry (4 of 4)

- In equilibrium,

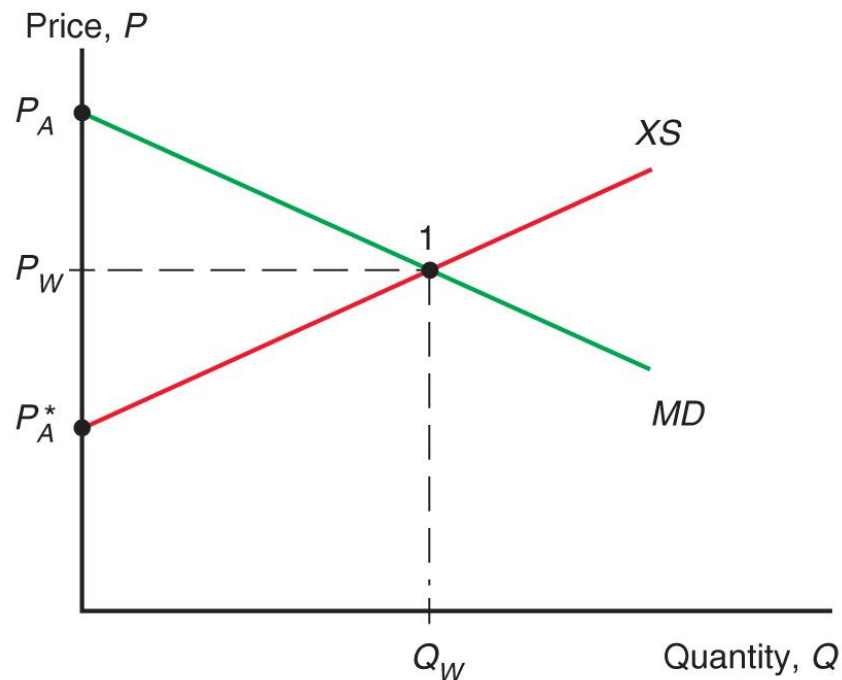
import demand = export supply,

home demand – home supply = foreign supply – foreign demand,

home demand + foreign demand = home supply + foreign supply,

world demand = world supply.

Figure 9.3 World Equilibrium



The equilibrium world price is where Home import demand (MD curve) equals Foreign export supply (XS curve).

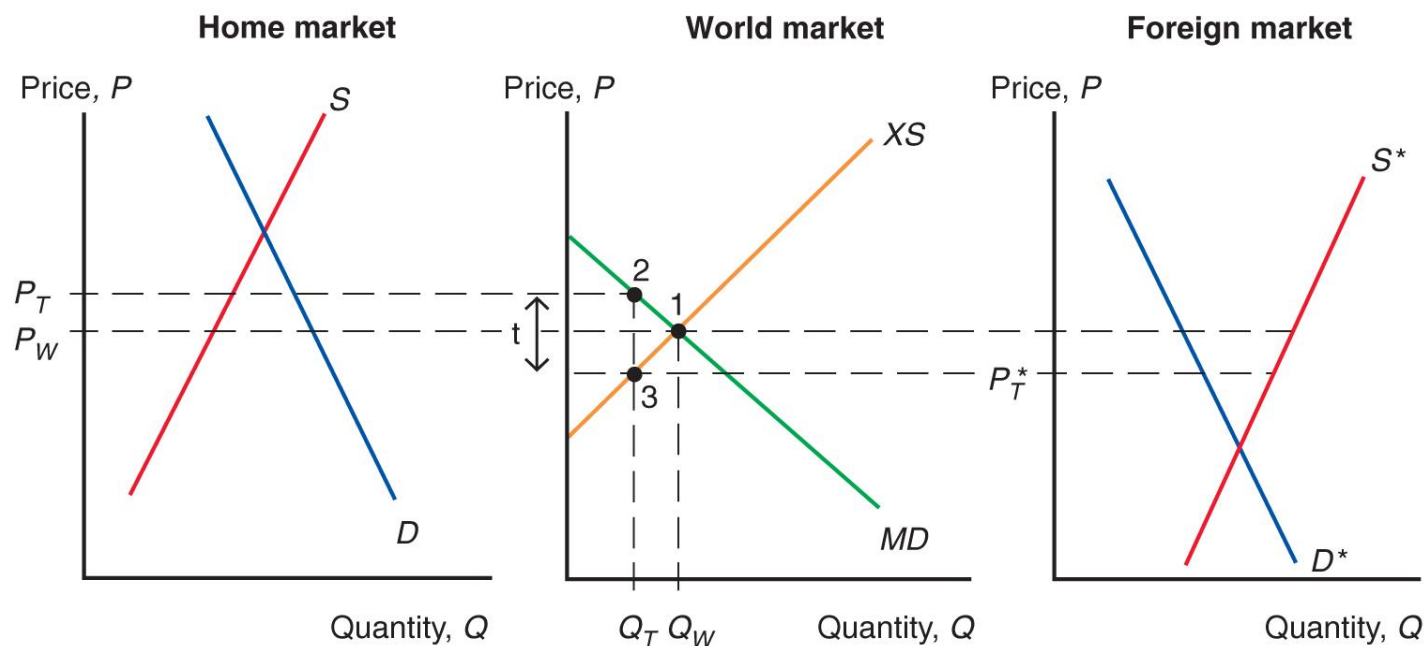
Effects of a Tariff (1 of 4)

- A tariff acts like a transportation cost, making sellers unwilling to ship goods unless the Home price exceeds the Foreign price by the amount of the tariff:

$$P_T - t = P_T^*$$

- A tariff makes the price rise in the Home market and fall in the Foreign market.

Figure 9.4 Effects of a Tariff



A tariff raises the price in Home while lowering the price in Foreign. The volume traded thus declines.

Effects of a Tariff (3 of 4)

- Because the price in the Foreign market falls from P_w under free trade to P_T with the tariff,
 - Home producers supply more and Home consumers demand less, so
 - the quantity of imports falls from Q_w under free trade to Q_T with the tariff.

Effects of a Tariff (2 of 4)

- Because the price in the Home market rises from P_w under free trade to P_T^* with the tariff,
 - Foreign producers supply less, and Foreign consumers demand more, so
 - the quantity of exports falls from Q_w to Q_T .

Effects of a Tariff (4 of 4)

- The quantity of Home imports demanded equals the quantity of Foreign exports supplied when

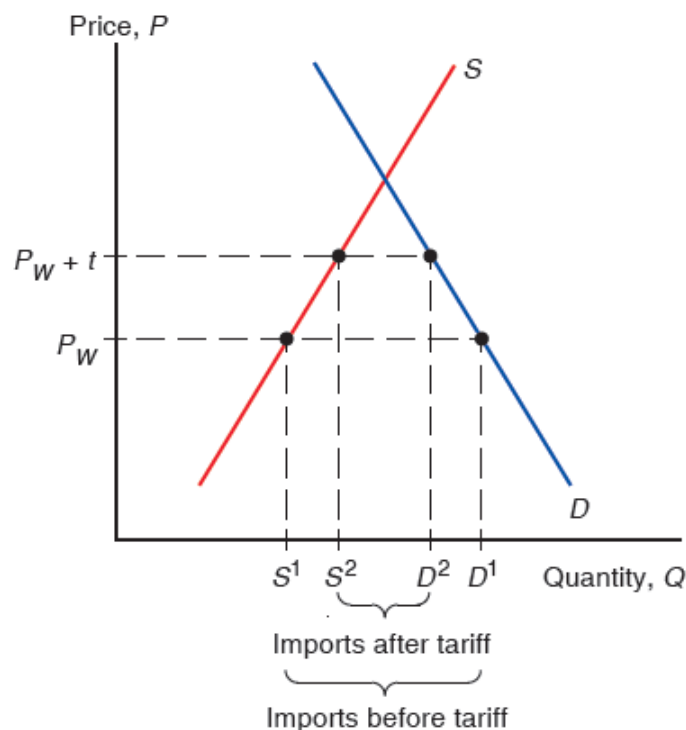
$$P_T - P_T^* = t$$

- The increase in the price in Home can be less than the amount of the tariff.
 - Part of the effect of the tariff causes the Foreign export price to decline.
 - But this effect is sometimes very small.

Effects of a Tariff in a Small Country

- When a country is “small,” it has no effect on the foreign (world) price because its demand is an insignificant part of world demand for the good.
 - The foreign price does not fall, but remains at P_w .
 - The price in the home market rises by the full amount of the tariff, to $P_T = P_w + t$.

Figure 9.5 A Tariff in a Small Country



When a country is small, a tariff it imposes cannot lower the foreign price of the good it imports. As a result, the price of the import rises from P_W to $P_W + t$ and the quantity of imports demanded falls from $D_1 - S_1$ to $D_2 - S_2$.

Measuring the Amount of Protection (1 of 3)

- The **effective rate of protection** measures how much protection a tariff (or other trade policy) provides.
 - It represents the change in value that firms in an industry add to the production process when trade policy changes, which depends on the change in prices the trade policy causes.
- Effective rates of protection often differ from tariff rates because tariffs affect sectors other than the protected sector, causing indirect effects on the prices and value added for the protected sector.

Measuring the Amount of Protection (2 of 3)

- For example, suppose that automobiles sell in world markets for \$8,000, and they are made from factors of production worth \$6,000.
 - The value added of the production process is $\$8,000 - \$6,000$.
- Suppose that a country puts a 25% tariff on imported autos so that home auto assembly firms can now charge up to \$10,000 instead of \$8,000.

Measuring the Amount of Protection (3 of 3)

- The effective rate of protection for home auto assembly firms is the change in value added:

$$\frac{(\$4,000 - \$2,000)}{\$2,000} = 100\%$$

- In this case, the effective rate of protection is greater than the tariff rate.

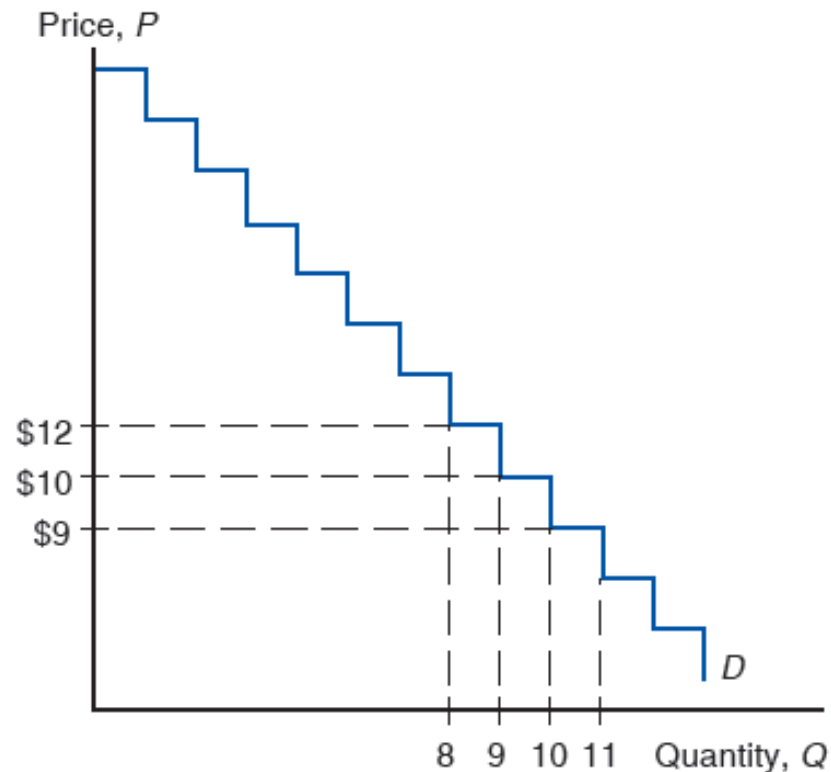
Costs and Benefits of Tariffs

- A tariff raises the price of a good in the importing country, so it hurts consumers and benefits producers there.
- In addition, the government gains tariff revenue.
- How to measure these costs and benefits?
- Use the concepts of consumer surplus and producer surplus.

Consumer and Producer Surplus (1 of 2)

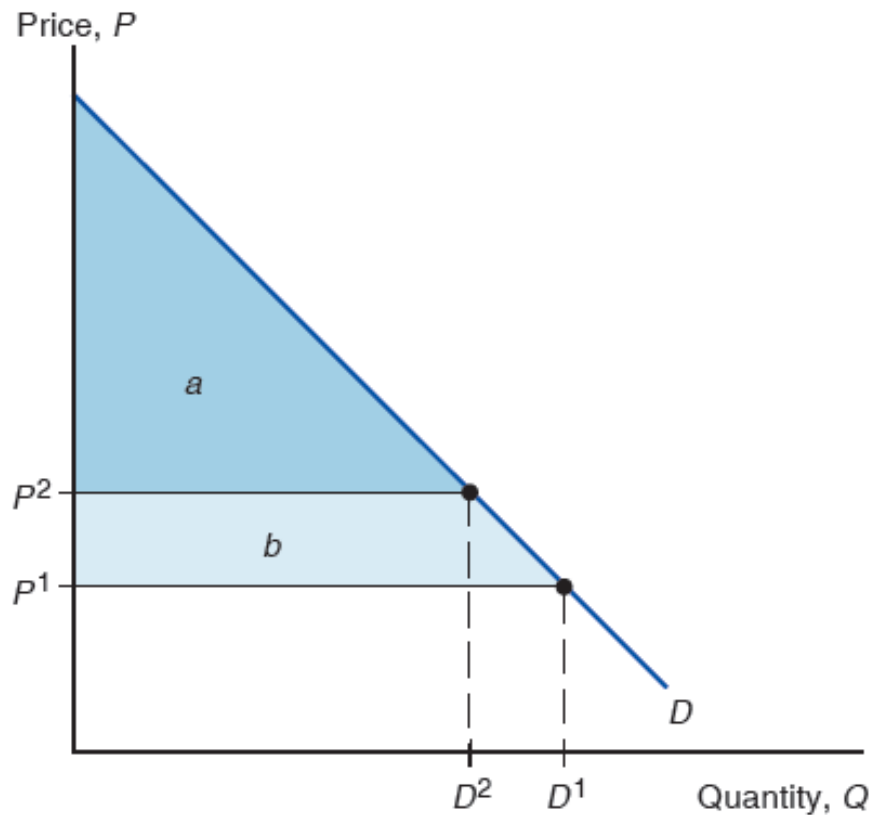
- **Consumer surplus** measures the amount that consumers gain from purchases by computing the difference in the price actually paid from the maximum price they would be willing to pay for each unit consumed.
 - When price increases, the quantity demanded decreases as well as the consumer surplus.

Figure 9.6 Deriving Consumer Surplus from the Demand Curve



Consumer surplus on each unit sold is the difference between the actual price and what consumers would have been willing to pay.

Figure 9.7 Geometry of Consumer Surplus

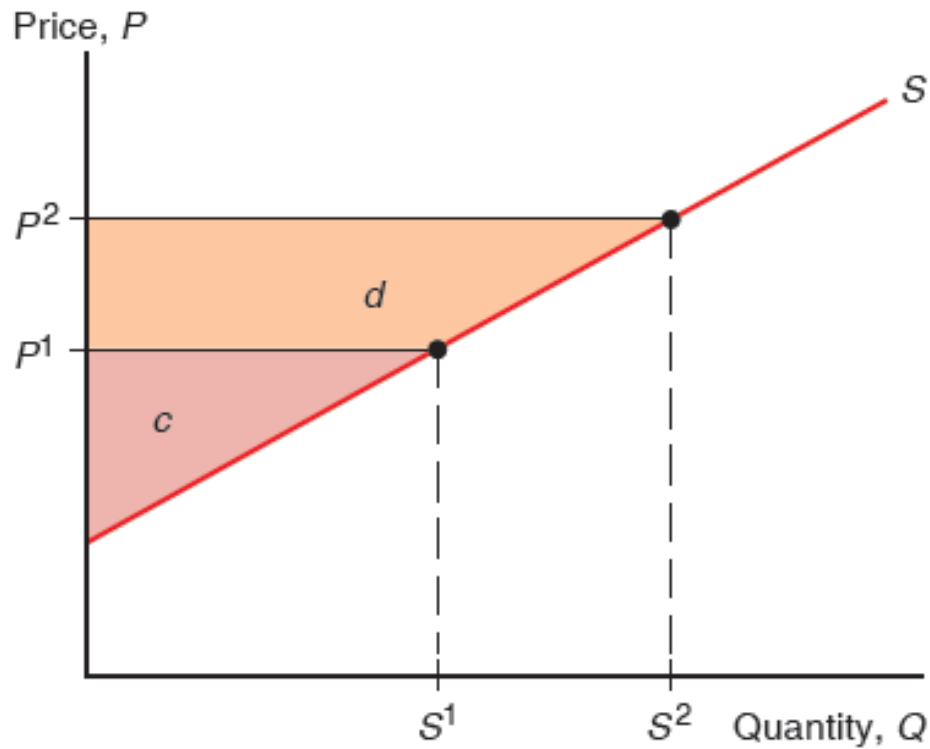


Consumer surplus is equal to the area under the demand curve and above the price.

Consumer and Producer Surplus (2 of 2)

- **Producer surplus** measures the amount that producers gain from sales by computing the difference in the price received from the minimum price at which they would be willing to sell.
 - When price increases, the quantity supplied increases as well as the producer surplus.

Figure 9.8 Geometry of Producer Surplus



Producer surplus is equal to the area above the supply curve and below the price.

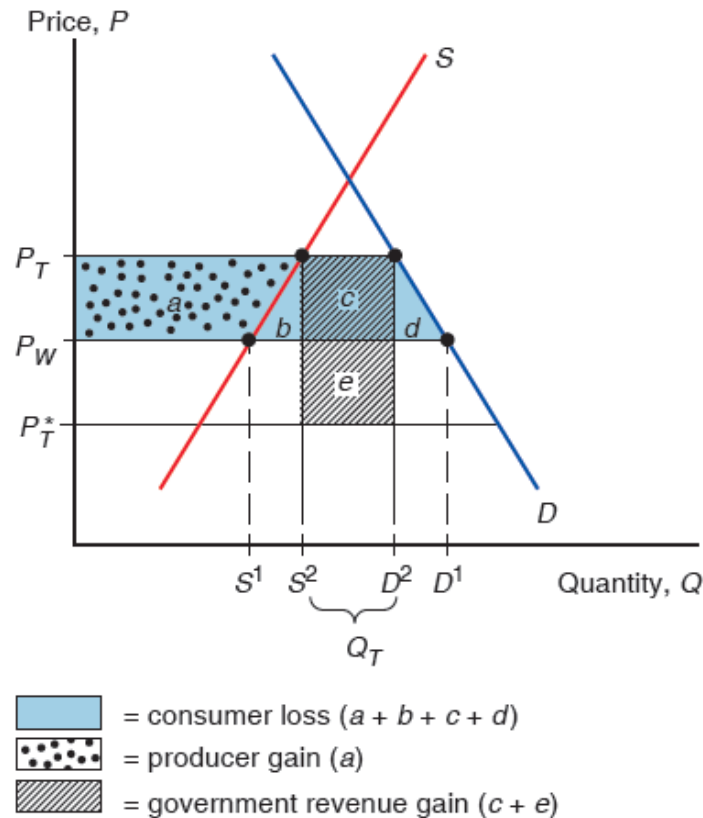
Measuring the Costs and Benefits of Tariffs (1 of 4)

- A tariff raises the price in the importing country:
 - consumer surplus decreases (consumers worse off)
 - producer surplus increases (producers better off).
 - the government collects tariff revenue equal to the tariff rate times the quantity of imports with the tariff.

$$tQ_T = (P_T - P_T^*)(D_2 - S_2)$$

- Change in welfare due to the tariff is $e - (b + d)$.

Figure 9.9 Costs and Benefits of a Tariff for the Importing Country



The costs and benefits to different groups can be represented as sums of the five areas a , b , c , d , and e .

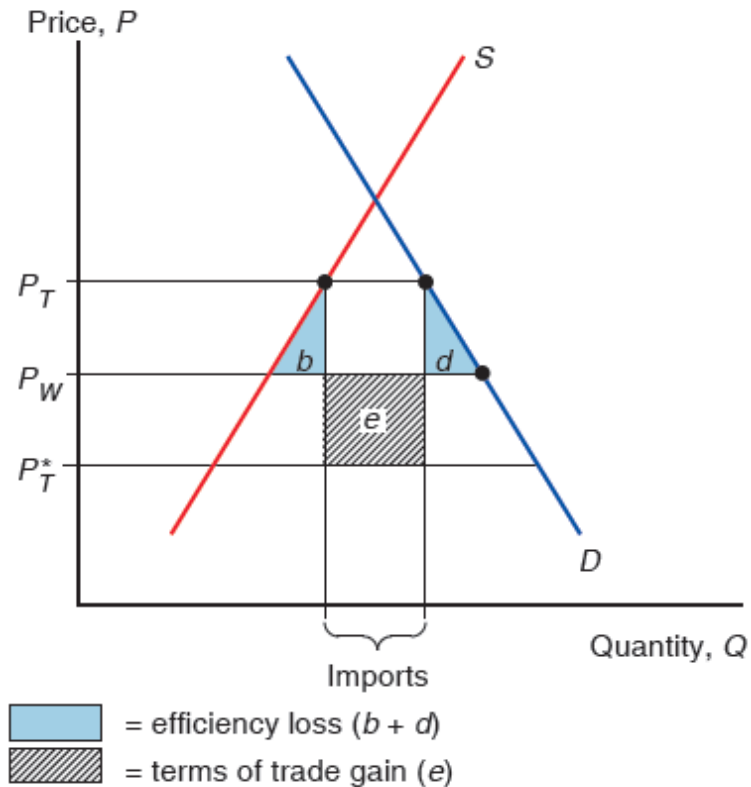
Measuring the Costs and Benefits of Tariffs (2 of 4)

- For a “large” country, whose imports and exports affect world prices, the welfare effect of a tariff is ambiguous.
- The triangles b and d represent the **efficiency loss**.
 - The tariff distorts production and consumption decisions: producers produce too much and consumers consume too little.
- The rectangle e represents the **terms of trade gain**.
 - The tariff lowers the Foreign price, allowing Home to buy its imports cheaper.

Measuring the Costs and Benefits of Tariffs (3 of 4)

- Part of government revenue (rectangle *e*) represents the terms of trade gain, and part (rectangle *c*) represents some of the loss in consumer surplus.
 - The government gains at the expense of consumers and foreigners.
- If the terms of trade gain exceed the efficiency loss, then national welfare will increase under a tariff, at the expense of foreign countries.
 - However, foreign countries are apt to retaliate.

Figure 9.10 Net Welfare Effects of a Tariff



The colored triangles represent efficiency losses, while the rectangle represents a terms of trade gain.

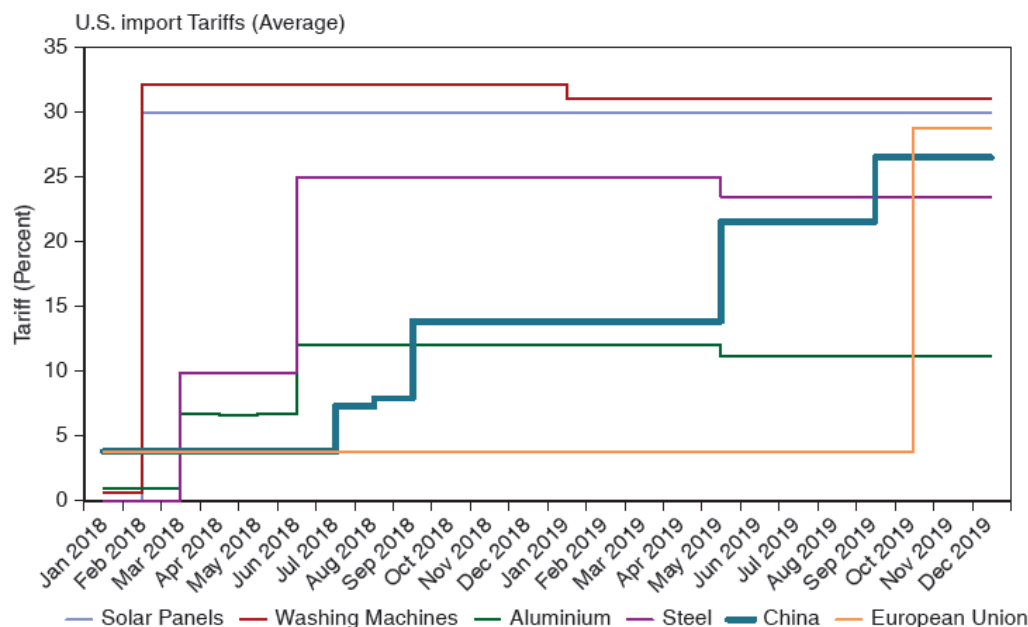
Measuring the Costs and Benefits of Tariffs (4 of 4)

- Tariffs can lead trading partners to retaliate with their own tariffs, thus hurting exporters in the country that first adopted the tariff.
- Tariffs can be hard to remove and large tariffs may induce producers to engage in wasteful activities to avoid paying tariffs.
 - Ford and Subaru install (then later remove) seats in vans and pickups trucks to avoid U.S. tariff on imports of light commercial trucks.

Winners and Losers of the Trump Trade War (1 of 3)

- Looking at the timeline of average tariffs imposed by the Trump administration 2018–2019:
 - Some tariffs on specific goods (solar panels, washing machines, steel, and aluminum) were imposed on a large set of exporting countries.
 - Some tariffs were targeted at specific countries (mostly China but also the European Union later on) and covered many different goods.
 - By 2019, over 66% of Chinese exports to the United States were hit by the tariffs shown in Figure 9-11.

Figure 9.11 Average Tariffs on U.S. Imports



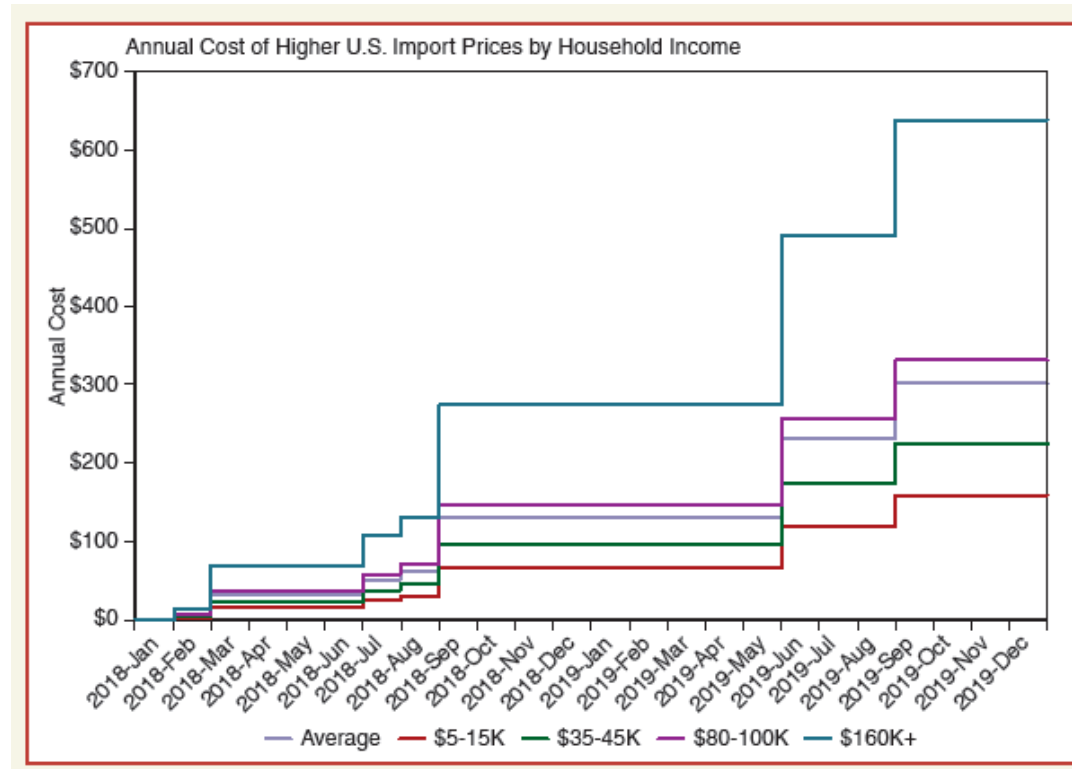
Over the 2018–2019 period, the Trump administration drastically increased tariffs targeted both at specific goods as well as specific trading partners.

Source: 2019 update of Pablo D. Fajgelbaum, Pinelopi K. Goldberg, Patrick J. Kennedy, and Amit K. Khandelwal, “The Return to Protectionism,” **The Quarterly Journal of Economics** 135, no. 1 (February 1, 2020), pp. 1–55.

Winners and Losers of the Trump Trade War (2 of 3)

- How have those tariff increases affected the prices of goods consumed in the United States?
- Any terms-of-trade gains were negligible—foreign exporters did not lower their prices to absorb hardly any of the tariffs.
- Almost the entire brunt of the tariffs has been borne by U.S. consumers and firms buying intermediate goods.
- The higher import prices due to the tariffs represented a substantial annual cost increase of \$300-900 for U.S. households.

Figure 9.12 Annual Cost of Higher Prices by Household Income



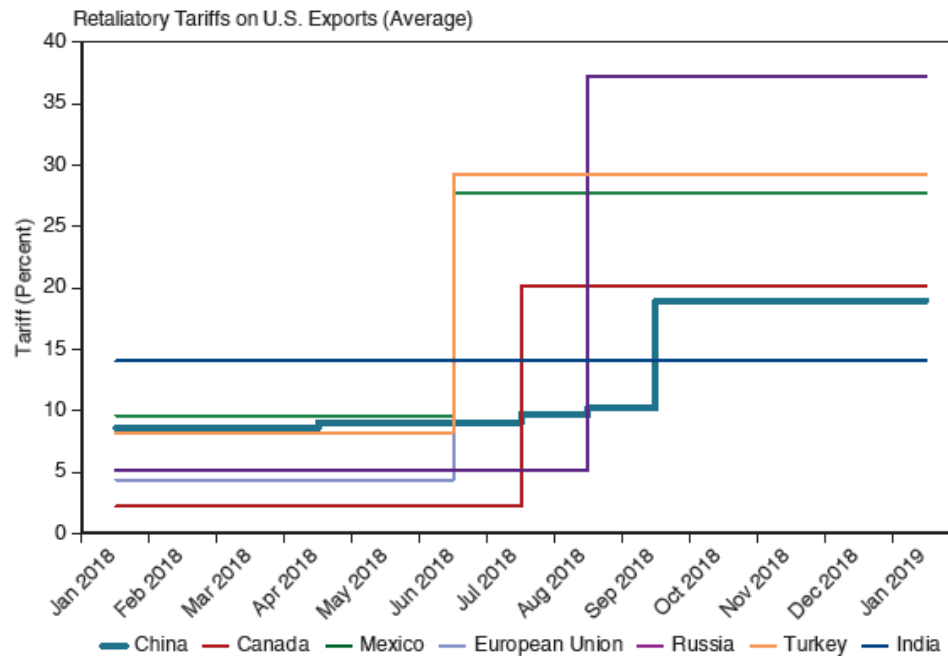
The higher prices for imported goods due to the new tariffs represented a substantial annual cost increase for U.S. households.

Source: Kirill Borusyak and Xavier Jaravel, "The Distributional Effects of Trade: Theory and Evidence from the United States," SSRN Scholarly Paper, October 6, 2018.

Winners and Losers of the Trump Trade War (3 of 3)

- The new tariffs imposed by the United States inevitably led to retaliation by its trading partners. Looking at the progression of those retaliatory tariffs (averaged over the set of targeted goods) over time:
 - China responded with higher tariffs on par with the higher tariffs imposed by the United States.
 - Major steel and aluminum exporters to the United States responded with tariffs on various U.S. imports.
 - All those retaliatory tariffs overwhelmingly targeted agricultural products. The Chinese tariffs covered virtually all U.S. agricultural exports.

Figure 9.13 Average Retaliatory Tariffs on U.S. Exports



All of the United States' major trading partners responded to the tariffs on their exports with tariffs of their own targeting U.S. exports.

Source: 2019 update of Pablo D. Fajgelbaum, Pinelopi K. Goldberg, Patrick J. Kennedy, and Amit K. Khandelwal, "The Return to Protectionism," **The Quarterly Journal of Economics** 135, no. 1 (February 1, 2020), pp. 1–55.

Tariffs for the Long Haul

- Tariffs can induce producers to behave in creative —though ultimately wasteful—ways in order to avoid them.
- Ford wants to avoid paying 25% tariff on light commercial trucks when imports its small commercial van from Europe.
 - Ford installs rear windows, rear seats, and seat belts prior to shipping the vehicles to the United States in an attempt to be classified as passenger vehicles (and pay 2.5% tariff).
 - Upon arrival, process reversed before delivery to dealers.

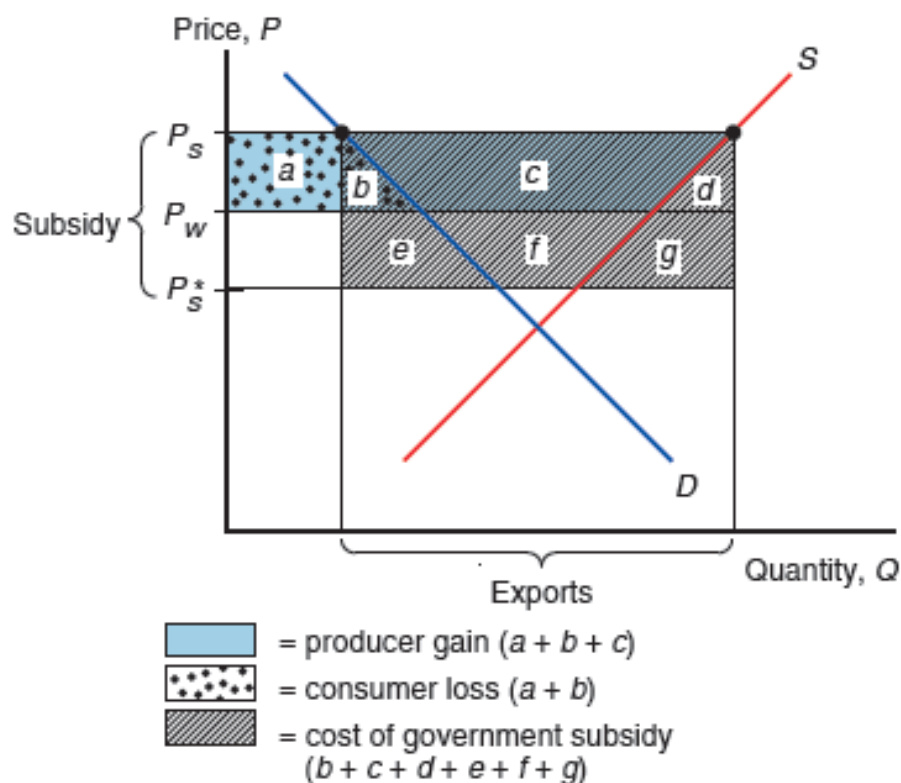
Export Subsidy (1 of 3)

- An export subsidy can also be **specific or ad valorem**:
 - A specific subsidy is a payment per unit exported.
 - An ad valorem subsidy is a payment as a proportion of the value exported.
- An export subsidy raises the price in the exporting country, decreasing its consumer surplus (consumers worse off) and increasing its producer surplus (producers better off).

Export Subsidy (2 of 3)

- Also, government revenue falls due to paying $S X_S^*$ for the export subsidy.
- An export subsidy lowers the price paid in importing countries $P_S^* = P_S - S$.
- In contrast to a tariff, an export subsidy worsens the terms of trade by lowering the price of exports in world markets.

Figure 9.14 Effects of an Export Subsidy



An export subsidy raises prices in the exporting country while lowering them in the importing country.

Export Subsidy (3 of 3)

- An export subsidy damages national welfare.
- The triangles b and d represent the **efficiency loss**.
 - The export subsidy distorts production and consumption decisions: producers produce too much and consumers consume too little compared to the market outcome.
- The area $b + c + d + f + g$ represents the **cost of the subsidy paid by the government**.
 - The terms of trade **decrease**, because the price of exports falls.

Import Quota (1 of 2)

- An **import quota** is a restriction on the quantity of a good that may be imported.
- This restriction is usually enforced by issuing licenses or quota rights.
- A binding import quota will push up the price of the import because the quantity demanded will exceed the quantity supplied by Home producers and from imports.

Import Quota (2 of 2)

- When a quota instead of a tariff is used to restrict imports, the government receives no revenue.
 - Instead, the revenue from selling imports at high prices goes to quota license holders.
 - These extra revenues are called **quota rents**.

An Import Quota in Practice: U.S. Sugar (1 of 4)

- Imports of sugar into the United States limited and quota rights passed out to foreign governments.
- Price of sugar in the United States has remained well above world prices.
- U.S. consumers are hurt by more than U.S. producers benefit; foreigners earn quota rents.
 - Overall effect on national welfare negative.

An Import Quota in Practice: U.S. Sugar (2 of 4)

- Under NAFTA, Mexico's sugar exports were slowly exempted from the quota restrictions and the U.S. sugar price premium decreased to its lowest level in over 25 years.
- U.S. sugar producers complained, and the U.S. Commerce Department sharply reduced Mexican sugar imports.
 - Imposed a 64% anti-dumping tariff and then negotiated a suspension of the tariff in return for lower exports.

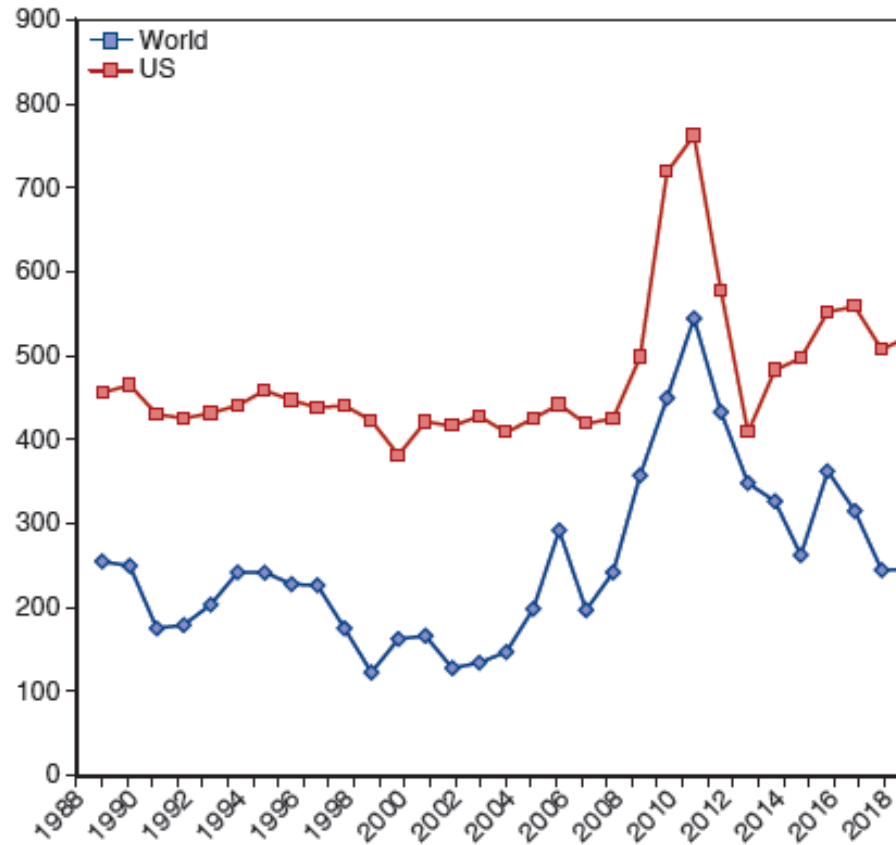
An Import Quota in Practice: U.S. Sugar (3 of 4)

- With access to the world sugar market successfully impeded, U.S. sugar prices have substantially risen again.
 - In 2015, that price was almost double the world price.
- For 2014, the sugar quota is estimated to:
 - cost consumers 3.5 billion (\$11 per person or \$30 for a typical household)
 - generate producer surplus losses for food producers who use refined sugar as an ingredient) of \$909 million
 - for a total cost estimate of \$4.4 billion
 - benefit sugar producers \$3.9 billion (mostly to refiners)

An Import Quota in Practice: U.S. Sugar (4 of 4)

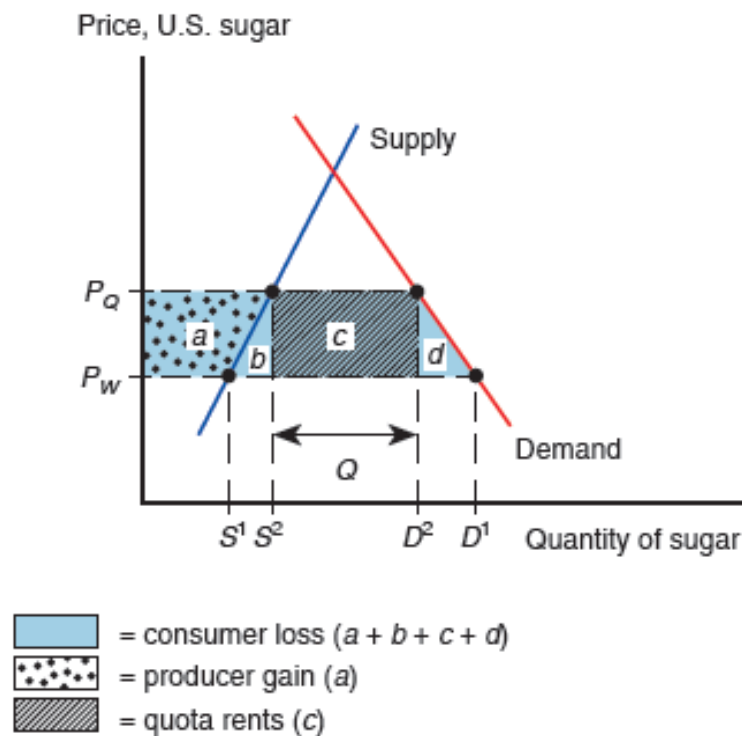
- Eliminating the sugar quota, by reducing the price of sugar in the United States, would generate 17,000–20,000 new jobs in producing foods containing sugar.
 - Far more than the 500–2,000 jobs that might be lost in the sugar industry.
 - Would turn the United States from a net importer to a net exporter of sugar-containing foods.
- The sugar producers are better lobbyists than the sugar-containing food sector so this protection has been extended.

Figure 9.15 U.S. and World Raw Sugar Prices, 1989–2019



Source: U.S. Department of Agriculture.

Figure 9.16 Effects of the U.S. Import Quota on Sugar



The quota level Q raises the price of sugar in the United States above the world price (from P_W to P_Q). The higher price associated with the quota induces an increase in U.S. sugar production (from S^1 to S^2) and a reduction in U.S. sugar consumption (from D^1 to D^2).

Voluntary Export Restraint

- A **voluntary export restraint** works like an import quota, except that the quota is imposed by the exporting country rather than the importing country.
- These restraints are usually requested by the importing country.
- The profits or rents from this policy are earned by foreign governments or foreign producers.
 - Foreigners sell a restricted quantity at an increased price.

A Voluntary Export Restraint in Practice (1 of 2)

- In 1979, sharp oil price increases caused the U.S. market to shift abruptly toward smaller cars.
- Japanese producers moved in to fill the increased demand faster than U.S. auto companies could come out with smaller, more fuel-efficient models.
- As the Japanese market share soared and U.S. output fell, strong political forces in the United States demanded protection.
- Rather than act unilaterally and risk creating a trade war, the U.S. government asked the Japanese government to limit its exports.

A Voluntary Export Restraint in Practice (2 of 2)

- The Japanese, fearing unilateral U.S. protectionist measures if they did not do so, agreed to limit their sales.
 - The first agreement, in 1981, limited Japanese exports to the United States to 1.68 million automobiles.
 - A revision raised that total to 1.85 million in 1984.
 - In 1985, the agreement was allowed to lapse.
- The price of Japanese cars in the United States rose, with the rent captured by Japanese firms.
- The total costs to the United States are estimated to have been \$3.2 billion in 1984.

Other Trade Policies

- Export credit subsidies
 - A subsidized loan to exporters
 - U.S. Export-Import Bank subsidizes loans to U.S. exporters.
- Government procurement
 - Government agencies are obligated to purchase from home suppliers, even when they charge higher prices (or have inferior quality) compared to foreign suppliers.
- Bureaucratic regulations (red tape)
 - Safety, health, quality, or customs regulations can act as a form of protection and trade restriction.

The Effects of Trade Policy: A Summary

- For each trade policy, the price rises in the Home country adopting the policy.
 - Home producers supply more and gain.
 - Home consumers demand less and lose.
- The world price falls when Home is a “large” country that affects world prices.
- Tariffs generate government revenue; export subsidies drain it; import quotas do not affect government revenue.
- All these trade policies create production and consumption distortions.

Table 9.1 Effects of Alternative Trade Policies

Policy	Tariff	Export Subsidy	Import Quota	Voluntary Export Restraint
Producer surplus	Increases	Increases	Increases	Increases
Consumer surplus	Falls	Falls	Falls	Falls
Government revenue	Increases	Falls (government spending rises)	No change (rents to license holders)	No change (rents to foreigners)
Overall national welfare	Ambiguous (falls for small country)	Falls	Ambiguous (falls for small country)	Falls